

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

First Semester of M. Sc. (Mathematics) Examination May 2018

MA723 : Linear Algebra

May 09, 2018, Wednesday

Time 10.00 a.m. to 10.45 a.m.

Maximum marks 20

Multiple Choice Questions

Instructions:

1. Tick the correct answer only in this sheet.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Each question carries one mark.
4. In this question paper, F denotes a field and $\mathbb{R}$ denotes the field of all real numbers.

- (1) Let V be a vector space of dimension 6 over F . Then the dimension of $\hat{V}$ is _____.
 (a) 6 (b) 36 (c) 0 (d) 1
- (2) Let V be a vector space of dimension 12 over F and W be a 4-dimensional subspace of V . Then the dimension of W° is _____, W° is the annihilator of W .
 (a) 4 (b) 12 (c) 8 (d) 3
- (3) Let V be a vector space over F and W and W_1 be subspaces of V such that $V = W + W_1$. If _____, then $V = W \oplus W_1$.
 (a) $V = W \cup W_1$ (b) $W + W_1 \subset V$
 (c) $W \cap W_1 \neq \{0\}$ (d) $W \cap W_1 = \{0\}$
- (4) Let V be a vector space over F and A, B be subspaces of V such that $A \cap B = \{0\}$. Then _____.
 (a) $\dim(A + B) = \dim(A) + \dim(B)$ (b) $\dim(A + B) = \dim(A) \dim(B)$
 (c) $\dim(A + B) = \dim(A) - \dim(B)$ (d) $\dim(A + B) = \dim(A)/\dim(B)$
- (5) Let $W = \{ (x_1, x_2, x_3, x_4, x_5) \in F^5 : x_1 + 2x_2 - 4x_3 + 7x_4 + 3x_5 = 0 \}$. Then $\dim W =$ _____.
 (a) 1 (b) 5 (c) 6 (d) 4
- (6) Let V be a vector space over F and $T \in A(V)$, $0 \neq T \neq I$ be such that $T^2 = T$. Then _____.
 (a) $P(x) = 1 - x^2$ is the minimal polynomial for T (b) T is nilpotent
 (c) 0 and 1 are characteristic roots of T (d) T is regular
- (7) Let V be a finite dimensional vector space over F and $S \in A(V)$ be one to one. Then S is _____.
 (a) on to (b) not onto (c) nilpotent (d) singular
- (8) Let V be a vector space over F and $T \in A(V)$ be nilpotent. Then _____.
 (a) T is regular (b) $T + I$ is singular
 (c) $T + T^2$ is regular (d) $T + I$ is regular
- (9) Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $T(x_1, x_2, x_3) = (x_3, x_2, 0)$, $(x_1, x_2, x_3) \in \mathbb{R}^3$. Then $\text{rank}(T) =$ _____.
 (a) 2 (b) 3 (c) 0 (d) 5
- (10) Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $T(x_1, x_2, x_3) = (0, x_1, x_2)$, $(x_1, x_2, x_3) \in \mathbb{R}^3$. Then the minimal polynomial for T is $p(x) =$ _____.
 (a) $1 - x^2$ (b) $1 + x^2$ (c) x^3 (d) x^2
- (11) Let V be a 5-dimensional vector space over F and $0 \neq T \in A(V)$. Suppose k is the smallest integer such that $T^k = 0$. Then _____.
 (a) $k = 0$ (b) $k = 1$ (c) $k > 5$ (d) $k \leq 5$
- (12) Let A and B be square matrices of order $n > 1$ over F be nilpotent. Then AB is nilpotent if _____.
 (a) $AB = BA$ (b) $AB \neq BA$ (c) $AB = A + B$ (d) none of these
- (13) Let V be a vector space of dimension 5 over F and $0 \neq T \in A(V)$ be nilpotent. Then the invariants of T may be _____.
 (a) 3, 2 (b) 4, 2 (c) 1, 1, 1, 1, 1 (d) 2, 3
- (14) Define $T : F^4 \rightarrow F^4$ by $T(x_1, x_2, x_3, x_4) = (0, x_1, x_2, 0)$, $(x_1, x_2, x_3, x_4) \in F^4$. Then the invariants of T are _____.
 (a) 4 (b) 3, 1 (c) 2, 2 (d) 2, 1, 1
- (15) Let V be a four dimensional vector space over F and $T \in A(V)$ be nilpotent and $\lambda \in F$ be a characteristic root of T then _____.
 (a) $\lambda = 4$ (b) $\lambda \neq 0$ (c) $\lambda = 0$ (d) $\lambda = 1$

- (16) Let V be a vector space over F and $S, T \in A(V)$ such that $\text{tr}(S) = 0$. Then _____.
- (a) $\text{Tr}(ST) = 0$ (b) $\text{Tr}(T+S) = \text{Tr}(T)$ (c) $\text{Tr}(S+T) = \text{Tr}(ST)$ (d) $\text{Tr}(TS) = 0$.
- (17) Let A and B be square matrices of order $n > 1$ over F . Then _____.
- (a) $\det \alpha A = \alpha (\det A)$, $\alpha \in F$ (b) $\det (A + B) = \det (A) + \det(B)$
(c) $\det (A^3) = (\det A)^3$ (d) $\det AB = (\det A) + \det (B)$
- (18) Let V be a finite dimensional vector space over F and $T \in A(V)$ such that $T^2 + 2T - 3I = 0$. Then _____.
- (a) $\det(T) = 0$ (b) $\text{Tr}(T) = 0$ (c) T is regular (d) T is singular
- (19) Let A be a square matrix of order 3 over F such that $\det(A) = 1$. Then _____.
- (a) $\det(A^{-1}) = 0$ (b) $\det(A^{-1}) = 1/3$ (c) $\det(A^{-1}) = 3$ (d) $\det(A^{-1}) = 1$
- (20) Consider a quadratic equation in three variables with a nonzero constant term whose inertia is $(1, 1, 1)$. Then the equation represents _____.
- (a) ellipsoid (b) hyperbolic cylinder
(c) hyperboloid of one sheet (d) elliptic cylinder

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First Semester of M. Sc. (Mathematics) Examination May 2018

MA723 : Linear Algebra

May 09, 2018, Wednesday

Time 10.45 a.m. to 01.00 p.m.

Maximum marks 50

Instructions:

1. Section I and Section II must be written in separate answer books.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Figures to the right indicate marks.
4. In this question paper, F denotes a field and $\mathbb{R}$ denotes the field of all real numbers.

SECTION - I

- Q-II (A) Attempt any TWO of the following: [8]
- (1) Let V be a vector space over F ; $T \in A(V)$ and $\lambda_1, \lambda_2, \dots, \lambda_k \in F$ be distinct characteristic roots of T and $v_1, v_2, \dots, v_k$ be characteristic vectors belonging to $\lambda_1, \lambda_2, \dots, \lambda_k$ respectively. Show that $v_1, v_2, \dots, v_k$ are linearly independent.
 - (2) Let V be a finite dimensional vector space over F ; $T \in A(V)$ and V_1, V_2 be subspaces of V such that V_1, V_2 are invariant under T and $V = V_1 \oplus V_2$. Let $T_i = T|_{V_i}$ and $p_i(x) \in F[x]$ be minimal polynomial for T_i , $i=1, 2$. Show that least common multiple of $p_1(x)$ and $p_2(x)$ is the minimal polynomial for T .
 - (3) Let V be a vector space over F . Giving all details, define a natural homomorphism $\psi : V \rightarrow \hat{V}$. Show that it is isomorphism. If $\dim(V) < \infty$, show that ψ is onto.
- (B) Attempt any SIX of the following: [12]
- (1) Let V be a finite dimensional vector space over F and W be a subspace of V . Show that there exists a subspace W_1 of V such that $V = W \oplus W_1$.
 - (2) Let V be a finite dimensional vector space over F ; W be a subspace of V . Show that $(W^\circ)^\circ = W$, where W° is the annihilator of W .
 - (3) Let V be a finite dimensional vector space over F and $T \in A(V)$. Show that T is regular iff T is onto.
 - (4) Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $T(x_1, x_2, x_3) = (2x_1 + x_2, 3x_1 + x_2, -9x_2 + x_3)$, $(x_1, x_2, x_3) \in \mathbb{R}^3$. Find the matrix of T w. r. t. the basis $v_1 = (1, 1, 1)$, $v_2 = (1, 1, 0)$, $v_3 = (1, 0, 0)$ of $\mathbb{R}^3$.
 - (5) Let V be a finite dimensional vector space over F and $S, T \in A(V)$. Show that $\text{rank}(ST) \leq \text{rank}(T)$.
 - (6) Let V be a finite dimensional vector space over F and $T \in A(V)$ be nilpotent having index of nilpotence k . Suppose $v \in V$ such that $T^{k-1}(v) \neq 0$. Show that $v, Tv, \dots, T^{k-1}(v)$ are linearly independent.
 - (7) Let V be a vector space over F and $T \in A(V)$ be nilpotent. Show that $I - T$ is regular. Find the inverse of $I - T$.
 - (8) Let A be square matrix of order $n > 1$ over F . If B is a matrix obtained from A by interchanging two rows of A . Then $\det(B) = -\det(A)$.
 - (9) Let V be a finite dimensional vector space over F and $T \in A(V)$ be such that all characteristic roots of T are in F . Show that $\text{trace of } T = \text{the sum of all characteristic roots of } T$.

SECTION-II

Q-III (A) Attempt any THREE of the following:

[18]

- (1) Let V be a finite dimensional vector space over F and W be a subspace of V . Show that W is finite dimensional and $\dim(V/W) = \dim(V) - \dim(W)$.
- (2) Let V be a finite dimensional vector space over F and W be a subspace of V . Show that $\dim(W^\circ) = \dim(V) - \dim(W)$, where W° is the annihilator of W .
- (3) Let V be a finite dimensional vector space over F and $T \in A(V)$ be such that all characteristic roots of T are in F . Show that there is a basis of V in which the matrix of T is triangular.
- (4) Let V be a finite dimensional vector space over F and $T \in A(V)$ be nilpotent. Show that invariants for T are unique.
- (5) Let A and B be square matrices of order $n > 1$ over F . Show that $\det(AB) = \det(A) \det(B)$.

(B) Attempt any THREE of the following:

[12]

- (1) Let V be a finite dimensional vector space over F ; $S, T \in A(V)$ and B be a basis of V . Show that the $m(ST) = m(S) m(T)$, where $m(S)$ is the matrix of S w. r. t. the basis B .
- (2) Let V be a finite dimensional vector space over F and $T \in A(V)$. Show that T is regular iff the constant term of the minimal polynomial is nonzero.
- (3) Diagonalize the matrix $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$, if it is diagonalizable.
- (4) Let F be a field of characteristic 0; V be a finite dimensional vector space over F ; and $S, T \in A(V)$ be such that $ST - TS$ commutes with S . Show that $ST - TS$ is nilpotent.
- (5) State and prove Cramer's rule.